Fourth-Order Perturbations of Pure-Weyl Black Holes

Exact Ricci–Bach Composition, Horizon Pairing, and Endpoint Nonselection

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Abstract

We ask whether familiar one-ended Lorentzian endpoint conditions isolate the Einstein sector of four-dimensional pure Weyl gravity around a black hole. We first classify the Bach-flat locus in a declared static Laurent ansatz, recover the Mannheim–Kazanas component, and derive a residual-basic horizon generator. On its nondegenerate static domain, the associated Lee–Wald Hamiltonian, Wald entropy, and signed surface-gravity temperature obey an exact first law at every simple horizon. We correct an important sheet distinction: on the complete Laurent locus the metric is Einstein precisely when $\gamma = 0$ and $w = 1$, whereas $\gamma = 0$ alone suffices only on the Mannheim–Kazanas component through $w = 1$.

On Schwarzschild, linearizing the Bach tensor gives an exact Ricci-carrier composition. It defines an exact sequence from Bach solutions to curvature solutions, not a canonical Einstein/additional direct sum. In the axial $\ell = 2$ carrier, $\psi_{ab} = \delta R_{ab}$ obeys a second-order Lichnerowicz-type equation. For nonzero real frequency its horizon system has a two-dimensional analytic ingoing ψ-space, so future-horizon regularity does not force $\psi = 0$. The action-derived Lee–Wald current makes the Einstein self-block isotropic for conjugate Regge–Wheeler waves. Controlled finite-series fixtures at three frequencies exhibit nonzero Einstein–additional cross pairing and nonzero additional-sector pairing; these fixture results are not promoted to a symbolic all-frequency theorem.

At infinity the certified characteristic polynomial is repeated. Until its Jordan structure, metric reconstruction, finite-flux falloff, and polar analogue are completed, we claim only endpoint nonselection: horizon analyticity together with the presently tested real-frequency leading asymptotics does not imply $\delta R_{ab} = 0$. This is not a no-go against all local causal truncations; the local Cauchy restriction $\psi|_\Sigma = \nabla_n \psi|_\Sigma = 0$ selects the Einstein kernel at the linear level. We make no claim about stability, quasinormal spectra, nonlinear closure, Hawking radiation, or quantum unitarity.

AI authorship and computational accountability

The named AI systems developed the programme, derivations, symbolic implementations, independent verification architecture, claim audits, and this manuscript. Asger Alstrup Palm commissioned and coordinated the work and is the corresponding human contact. Every certified statement below is bound to a machine-readable certificate and a separately implemented verifier. The cross-flux result uses controlled high-precision series evaluation with exact recurrence data and independent frequencies; it is reported as fixture-level evidence, not interval-certified nonvanishing. All local

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tensor identities and polynomial classifications are exact. The manuscript remains conditional on final human review.

Contents

1 The physical question

Fourth-order gravitational equations contain more local solutions than the Einstein equations. A common proposal is to retain the Einstein branch by a boundary condition. Around a one-ended Lorentzian black hole, a first test is:

> Do future-horizon analyticity and the same leading outer admissibility tests used for Einstein waves force the additional Ricci carrier to vanish?

The answer in the certified Schwarzschild axial $\ell = 2$ mode laboratory is no: the curvature carrier has regular ingoing horizon data, the leading real-frequency symbol does not separate it from the Einstein carrier, and controlled fixtures have nonzero action-derived horizon pairing. This is an endpoint-admissibility result, not a classification of all local differential boundary conditions. In particular, setting the Ricci carrier and its normal derivative to zero on a Cauchy surface is a local linear Einstein-sector restriction. Whether that restriction is natural, interaction-stable, or compatible with a physical asymptotic phase space is open.

1.1 Claims and dependency tags

The static family and local operator identities carry LOCAL-ALGEBRAIC. Harmonic branch, horizon, current, and asymptotic statements also carry REDUCED-MODE. No result in this manuscript is promoted to LORENTZIAN-CAUSAL: there is no complete exterior Green-hyperbolicity theorem, metric-falloff phase space, or classified set of local differential boundary operators.

2 Relation to existing branch-selection programmes

The static Bach-flat family itself is classical [2, 3]. Conformal-gravity thermodynamics with additional spin-two hair has also been studied in asymptotically AdS and Einstein–Weyl settings [14]. Our static contribution is therefore not a new family: it is the exact residual-basic normalization and simultaneous signed first-law identity within a declared local quotient. A physical asymptotic normalization has not yet been selected. In particular, Lü, Pang, Pope, and Vázquez-Poritz organize AdS thermodynamics with an independent massive-spin-two-hair conjugate pair. Our one-term identity instead follows after quotient-basic normalization on a restricted static parameter family. The different boundary ensemble, generator normalization, and independent variables mean that the two first laws are complementary, not contradictory.

Maldacena's Einstein-from-conformal construction selects an Einstein sector by Euclidean AdS or late-time de Sitter boundary data [8]. Critical- and logarithmic-gravity analyses similarly separate massless, massive, and generalized modes by asymptotic conditions [9, 13, 10, 11, 12]. The null Einstein self-pairing and nonzero cross pairing found below should be read in that established degenerate-operator context, rather than as the first appearance of null massless modes paired with generalized solutions. Our different question is whether one-ended Lorentzian Schwarzschild endpoint tests force the Ricci carrier to vanish.

Using δR_{ab} to expose higher-derivative spin-two modes is also a known strategy in fourth-order gravity [15]. Previous Weyl-black-hole quasinormal analyses often deliberately obtained a second-order master equation by conformal transport rather than retaining the full fourth-order Bach solution space [16]. Effective-Einstein perturbations of conformally related nonsingular metrics [17] and scalar or electromagnetic test-field spectra [18] are different problems: test-field stability is not stability of the dynamical fourth-order Bach sector. Contemporary quadratic-gravity ringdown work finds both ordinary and additional spin-two sectors [19], while time-domain quadratic-gravity studies track massive-sector tails [20]. The prospective advance here is narrower: the exact pure-Weyl Ricci–Bach sequence, its action-derived Schwarzschild horizon pairing, and failure of the tested endpoint conditions to imply $\delta R_{ab} = 0$. The first-order-system analysis follows the general modified-gravity asymptotic methodology of [21]; in particular, that methodology makes the unresolved repeated-root Jordan audit an essential next step rather than a technical detail. Precise decay of the metric coefficients must ultimately be compared with the Schwarzschild Einstein benchmark [22], not inferred from bounded curvature carrier variables alone.

3 Pure Weyl gravity and its covariant current

We use signature $(-+++)$ and action

$$S_W[g] = \alpha \int_M \sqrt{-g}\, C_{abcd} C^{abcd}\, \mathrm{d}^4 x. \tag{1}$$

The field equation is the Bach equation $B_{ab} = 0$, with convention

$$B_{ab} = \nabla^c \nabla^d C_{acbd} + \frac{1}{2} R^{cd} C_{acbd}. \tag{2}$$

The covariant presymplectic current is obtained from the literal variation of the curvature-momentum potential of this action. Its normalization is fixed by the off-shell Green identity

$$\partial_t F^t(h_A, h_B) + \partial_r F^r(h_A, h_B) = 4\alpha \int_{S^2} \sqrt{q}\, \big(h_B^{ab} \delta B_{ab}[h_A] - h_A^{ab} \delta B_{ab}[h_B]\big). \tag{3}$$

Thus the current is conserved for pairs of linearized Bach solutions and is not a freely rescaled master-equation Wronskian.

4 Static spherical Bach-flat family

In Schwarzschild gauge,

$$\mathrm{d}s^2 = -B(r)\mathrm{d}t^2 + B(r)^{-1}\mathrm{d}r^2 + r^2 \mathrm{d}\Omega_2^2, \tag{4}$$

the declared Laurent ansatz is

$$B(r) = w - \frac{u}{r} + \gamma r - k r^2 + \frac{c_2}{r^2} + c_3 r^3. \tag{5}$$

Theorem 4.1 (Complete Bach-flat locus in the Laurent ansatz)**.** *The metric is Bach-flat if and only if*

$$c_2 = c_3 = 0, \qquad w^2 + 3u\gamma = 1. \tag{6}$$

The Mannheim–Kazanas parametrization is recovered by

$$w = 1 - 3\beta\gamma, \qquad u = \beta(2 - 3\beta\gamma).$$

On the complete Laurent locus the metric is Einstein if and only if

$$\gamma = 0, \qquad w = 1. \tag{7}$$

On the Mannheim–Kazanas component through $w = 1$, this reduces to the single condition $\gamma = 0$.

Completeness here refers only to the Laurent ansatz. The certificate does not claim a new proof of the global static spherical classification. The sheet qualification in (7) follows already from the scalar curvature

$$R = 12k - \frac{6\gamma}{r} + \frac{2(1-w)}{r^2}. \tag{8}$$

In particular, the allowed point $(\gamma, w) = (0, -1)$ is not Einstein. The original static certificate's phrase "Einstein iff $\gamma = 0$" is therefore used only on its explicitly parametrized Mannheim–Kazanas sheet; the manuscript does not propagate it to the complementary Laurent sheet.

The form-preserving local Diff $\times$Weyl transformations act as dilation and translation on the cubic

$$Q(x) = -ux^3 + wx^2 + \gamma x - k, \qquad x = r^{-1}.$$

Their orbit rank is two. A single continuous invariant may be taken as

$$J = u^2\, \mathrm{disc}(Q),$$

while $\mathrm{disc}(Q) = 0$ is the degenerate-horizon locus. The conformal translation may fail globally where its Weyl factor vanishes; it is used only as a local orbit statement.

An exact non-Einstein three-horizon fixture is

$$(\beta, \gamma, k) = \left(\frac{3}{2}, \frac{12}{19}, \frac{1}{19}\right), \qquad rB(r) = -\frac{1}{19}(r-1)(r-3)(r-8).$$

All curvature invariants are finite at the simple horizons. In ingoing Eddington–Finkelstein coordinates,

$$\mathrm{d}s^2 = -B(r)\mathrm{d}v^2 + 2\mathrm{d}v\mathrm{d}r + r^2\mathrm{d}\Omega_2^2,$$

the metric and Bach equation are regular there.

5 Residual-basic energy and the static first law

For the chart generator ∂_t, the bare static Lee–Wald one-form is not integrable. The obstruction disappears when the generator is normalized consistently with the residual quotient.

Theorem 5.1 (Normalized static generator)**.** *On a connected component with $u \neq 0$, choose the oriented generator*

$$\chi = u\,\partial_t, \qquad u = \beta(2 - 3\beta\gamma). \tag{9}$$

Residual basicness and dilation weight determine the general normalization as $uf(J)\partial_t$. We select $f(J) = 1$. Its Hamiltonian is

$$H = -16\pi\alpha\beta^2 D_2, \qquad D_2 = 9\beta^2\gamma^2 k - \beta\gamma^3 - 12\beta\gamma k + \gamma^2 + 4k. \tag{10}$$

At every simple horizon r_h, the Wald entropy

$$S_h = 64\pi^2\alpha\beta\frac{2 - 3\beta\gamma + \gamma r_h}{r_h} \tag{11}$$

and normalized temperature

$$T_h = \frac{uB'(r_h)}{4\pi} \tag{12}$$

satisfy

$$\mathrm{d}H = T_h\,\mathrm{d}S_h \tag{13}$$

identically modulo $B(r_h) = 0$.

Here T_h is the *signed* surface-gravity temperature appropriate to the oriented first-law identity; the positive Hawking temperature would use $|uB'(r_h)|/(4\pi)$. On a component where $u < 0$, future-directedness uses $-\chi$, flipping both H and T_h and preserving the first law. The locus $u = 0$ is excluded because this normalization degenerates. For another allowed $f(J)$, one has $T_f = f(J)T_h$ and a reparametrized Hamiltonian satisfying $\mathrm{d}H_f = f(J)\mathrm{d}H$; the first law remains true. Thus residual basicness fixes a class, not a unique physical normalization in the absence of an asymptotic clock.

This is a theorem on the static parameter slice, not yet a general physical-process first law. Its linear dynamical extension nevertheless passes a strong gauge audit: arbitrary time-dependent spherical Weyl rescalings and diffeomorphisms have exactly zero charge and flux, the entropy is conformally invariant on the symbolic family, and the normalized generator extension is unique at linear order under the declared boundary scalar rule.

6 Ricci–Bach composition and the axial exact sequence

The parity-specific formulas are restrictions of one Ricci-flat identity.

Proposition 6.1 (Linearized Bach tensor on a Ricci-flat background)**.** *For a perturbation of a four-dimensional Ricci-flat metric,*

$$\delta B_{ab} = \frac{1}{2}\Box(\delta R_{ab}) + C_{acbd}(\delta R)^{cd} - \frac{1}{6}\nabla_a\nabla_b\delta R - \frac{1}{12}g_{ab}\Box\delta R. \tag{14}$$

Proof. Rewrite the four-dimensional Bach tensor in Ricci variables using the contracted Bianchi identity, then retain the terms linear in Ricci curvature about $R_{ab} = 0$. Background derivatives and the curvature commutator give the displayed Weyl action. The repository verifies (14) independently, component by component, on the complete axial and polar $\ell = 2$ Regge–Wheeler-gauge carriers. That computer check is carrier-complete but is not presented as a machine proof for arbitrary harmonic content. □

We now specialize the background to Schwarzschild and use axial Regge–Wheeler gauge,

$$h_{t\phi} = h_0(t,r)S_{\ell m}, \qquad h_{r\phi} = h_1(t,r)S_{\ell m}.$$

The $\ell = 2$ linearized Bach tensor is fourth order, trace-free, and obeys the exact linearized Noether identity. Eliminating the constraint on the Einstein subbranch reproduces the Regge–Wheeler equation for $\Psi = Bh_1/r$,

$$B(B\Psi')' + (\omega^2 - V_{\mathrm{RW}})\Psi = 0, \qquad V_{\mathrm{RW}} = B\left(\frac{6}{r^2} - \frac{6m}{r^3}\right). \tag{15}$$

Corollary 6.2 (Axial composition)**.** *For the certified axial carrier, parity gives $\delta R = 0$, so*

$$\boxed{\delta B_{ab} = \frac{1}{2}\Box(\delta R_{ab}) + C_{acbd}(\delta R)^{cd}.} \tag{16}$$

Consequently the Einstein kernel $\delta R_{ab} = 0$ injects into the Bach kernel, while its curvature image obeys

$$\frac{1}{2}\Box\psi_{ab} + C_{acbd}\psi^{cd} = 0, \qquad \psi_{ab} := \delta R_{ab}. \tag{17}$$

Let $\mathcal{C}_{\mathrm{ax}}$ be the fixed axial carrier and define

$$\mathcal{H}_B = \{h \in \mathcal{C}_{\mathrm{ax}} : \delta B[h] = 0\}, \qquad \mathcal{H}_E = \{h \in \mathcal{H}_B : \delta R[h] = 0\}, \tag{18}$$

$$\mathcal{H}_R = \ker\left(\tfrac{1}{2}\Box + C\circ\right) \cap \operatorname{im}(\delta R|_{\mathcal{C}_{\mathrm{ax}}}). \tag{19}$$

Then the invariant statement supplied by the composition is the exact sequence

$$0 \longrightarrow \mathcal{H}_E \longrightarrow \mathcal{H}_B \xrightarrow{\delta R} \mathcal{H}_R \longrightarrow 0. \tag{20}$$

It does *not* supply a canonical direct sum. A metric lift h_X of $\psi \in \mathcal{H}_R$, when it exists in the chosen domain, may be shifted by any $h_E \in \mathcal{H}_E$. Global lift existence, boundary conditions, and a gauge-independent splitting must therefore be proved separately. In particular, “additional branch” below refers either to the curvature image $\mathcal{H}_R$ or to an explicitly constructed lift, never to an assumed canonical complement.

The identity is componentwise and off shell in the perturbation. It does not extend in this two-term form to the certified non-Einstein static fixture. Thus the clean composition is a Ricci-flat-background theorem.

7 The Ricci carrier reaches the future horizon

In ingoing coordinates (v, r), Fourier modes are written $e^{i\omega v}$. The extra carrier's divergence constraint solves its third component polynomially; the apparent Schwarzschild-coordinate pole is a chart artifact.

Theorem 7.1 (Analytic ingoing curvature family)**.** *For axial $\ell = 2$ and $\omega \in \mathbb{R} \setminus \{0\}$, the future horizon $r = 2m$ is a regular singular point of the extra first-order radial system. Its residue spectrum is*

$$\{0, 0, -4im\omega, -2 - 4im\omega\}, \tag{21}$$

and the zero eigenspace has dimension two. Hence there is a two-parameter family of ψ-solutions analytic at the future horizon, with no leading logarithm at the zero exponent.

The restriction excludes $\omega = 0$, where the algebraic multiplicity of the zero residue exponent changes, and makes no claim at resonant complex frequencies. Most importantly, the theorem counts the curvature carrier, not a quotient of metric solutions. The currently certified dimensions are:

Space	Ingoing/analytic dimension	Status
Einstein Regge–Wheeler carrier	1	second-order control
Ricci carrier ψ	2	exact residue calculation
Full Bach metric solution space	not certified	reconstruction/domain open
Metric quotient by the Einstein kernel	not certified	lift and gauge open

Thus future-horizon regularity does not force $\psi = 0$. Explicit metric lifts are constructed in the flux fixtures below, but the exact sequence (20) has not yet been split globally.

8 Action-derived horizon flux

Substituting two on-shell Regge–Wheeler solutions into the current (3) gives

$$F^r_{EE} = -\frac{192\pi\alpha(\omega_1^2 - \omega_2^2)}{5\omega_1\omega_2 r}\Psi_1\Psi_2. \tag{22}$$

For conjugate pairs $\omega_2 = \pm\omega_1$, this vanishes identically.

Theorem 8.1 (Exact Einstein isotropy)**.** *On Schwarzschild in axial $\ell = 2$, the pure-Weyl Lee–Wald pairing of an Einstein Regge–Wheeler mode with its conjugate vanishes exactly:*

$$F^r(E_\omega, \overline{E_\omega}) = 0. \tag{23}$$

Thus the Einstein line is isotropic in the presymplectic structure inherited from the pure C^2 action. This statement does not identify the Einstein symplectic form with the ordinary Einstein–Hilbert one.

Proposition 8.2 (Controlled horizon-pairing fixtures)**.** *For $m = 1$, axial $\ell = 2$, and conjugate-frequency Hermitian pairings, order-16 ingoing series give*

$$F^r(E_\omega, \overline{X_\omega}) \neq 0, \qquad F^r(X_\omega, \overline{X_\omega}) \neq 0 \tag{24}$$

at producer frequencies $\omega = 3/5, 2/7$; an independent verifier repeats the nonvanishing gates at $\omega = 1/2$. At the two stored frequencies the values, divided by $\pi\alpha$ and sampled at two interior radii, are approximately

ω	$F^r(X, \bar{X})/(\pi\alpha)$	$F^r(E, \bar{X})/(\pi\alpha)$
3/5	$-330.58i, -329.98i$	$-218.17 + 43.55i, -215.97 + 51.53i$
2/7	$-71.92i, -71.95i$	$-162.82 + 27.99i, -163.92 + 33.03i$

The exact Regge–Wheeler null identity is the in-run control. These data are controlled numerical fixtures: they prove neither symbolic nonvanishing for all real ω nor an interval-certified error bound.

For a chosen lift basis (E, X), the horizon block has the form

$$\Omega_h = \begin{pmatrix} 0 & a \\ -\bar{a} & b \end{pmatrix}. \tag{25}$$

Under the admissible lift change $X \mapsto X + cE$, the cross entry a is invariant while b changes. The robust fixture-level conclusions are therefore Einstein isotropy, nonzero cross pairing, and full rank of the tested two-dimensional block. The sign of iF^r_{XX} for one chosen lift is basis dependent and is not promoted to a branch signature or Hilbert norm.

The stored report also records a repaired calculation: an earlier flux evaluation dropped the Fourier factor $e^{i\omega t}$, causing all time derivatives to vanish silently. The exact Regge–Wheeler null control exposed the defect; all displayed fixtures use the repaired full-time current.

9 Asymptotic symbol and endpoint nonselection

At real frequency and large radius, the extra carrier has dispersion

$$(\lambda_r^2 - \omega^2)^2 = 0 \tag{26}$$

and growth exponents

$$\sigma \in \{\pm 2im\omega,\ \pm 2im\omega - 1\}. \tag{27}$$

The simple-root reading would identify Coulomb logarithmic phases with amplitudes of order r^0 and r^{-1}. But the characteristic is *double*. The polynomial alone does not determine geometric multiplicity. As in $(D^2 + \omega^2)^2 h = 0$, a Jordan chain may produce $r_* e^{\pm i\omega r_*}$ even though every characteristic exponent is oscillatory. The current certificate does not print the asymptotic first-order matrix, its Jordan form, or the reconstructed metric behavior. Its "no growing asymptotics" flag is therefore used only for the displayed curvature-carrier leading ansatz, not as an asymptotically flat metric theorem.

Theorem 9.1 (Endpoint regularity-and-leading-symbol nonselection)**.** *On the declared Schwarzschild axial $\ell = 2$, nonzero-real-frequency mode carrier, future-horizon analyticity does not imply $\delta R_{ab} = 0$. Moreover, the presently certified leading characteristic polynomial and formal exponents do not supply an outer condition that distinguishes the Ricci carrier from the Einstein characteristic. Hence these tested local endpoint diagnostics do not select the Einstein kernel.*

This deliberately does *not* say that every ordinary asymptotically flat radiation condition admits an additional metric mode. That stronger statement requires: (i) the full asymptotic Jordan decomposition; (ii) reconstruction of h from $\psi = \delta R[h]$; (iii) falloff in a regular tetrad or asymptotic gauge; and (iv) finite Lee–Wald flux at null infinity. Nor is there a no-go against local causal selection. Since ψ obeys a second-order hyperbolic equation, the local Cauchy condition

$$\psi|_\Sigma = 0, \qquad \nabla_n \psi|_\Sigma = 0 \tag{28}$$

propagates $\psi = 0$ and selects the linear Einstein kernel. The open physical questions are whether this restriction is preserved by nonlinear interactions and whether it follows from a differentiable asymptotic phase space rather than being imposed by hand.

10 Polar carrier: exact composition and open causal chain

For polar $\ell = 2$ Regge–Wheeler-gauge perturbations, the trace variation δR need not vanish. The axial formula acquires two universal trace terms.

Corollary 10.1 (Polar trace-coupled composition)**.** *On the certified polar $\ell = 2$ carrier, Proposition 6.1 gives*

$$\boxed{\delta B_{ab} = \frac{1}{2}\Box(\delta R_{ab}) + C_{acbd}(\delta R)^{cd} - \frac{1}{6}\nabla_a \nabla_b \delta R - \frac{1}{12} g_{ab} \Box \delta R.} \tag{29}$$

The Einstein polar kernel injects by $\delta R_{ab} = 0$, while the curvature image obeys a second-order trace-coupled Lichnerowicz system for $(\delta R_{ab}, \delta R)$.

This opens, but does not complete, the polar analogue of the axial chain. As in (20), composition yields an exact sequence onto the realized Ricci image; it does not by itself produce a canonical metric complement. The Zerilli benchmark, horizon indicial system, action-derived flux blocks, outer asymptotics, and causal disposition remain pending. No parity-complete black-hole truncation theorem is claimed until those steps close.

11 Result ledger

Question	Certified answer	Scope
Static Bach-flat black holes?	exact Laurent-family classification and three-horizon fixture; Einstein sheet corrected	local/static
Differentiable energy?	residual-basic normalized generator, exact static first law	$u \neq 0$; signed orientation; $f(J)$ freedom
Einstein axial waves present?	yes; exact Regge–Wheeler injection	Schwarzschild, axial $\ell = 2$
Additional axial content?	nonzero second-order Ricci carrier; no canonical metric split	same
Reach future horizon?	two analytic ingoing ψ amplitudes	nonzero-real-frequency carrier
Carry current?	Einstein self-block exactly null; mixed/additional blocks nonzero in fixtures	exact identity plus three frequencies
Selected by tested endpoints?	no: horizon and leading-symbol tests do not force $\psi = 0$	not a general causal no-go
Asymptotic metric radiation?	open	repeated-root Jordan, reconstruction, and finite flux missing
Polar composition?	exact trace-coupled identity	causal chain still open
Stable/ringing?	not established	complex frequencies open

12 What is not proved

The paper does not establish:

- a complete exterior initial-boundary well-posedness theorem;
- general ℓ, m, the completed polar chain, or a non-Einstein background Ricci-carrier composition;
- a canonical splitting of the Bach solution space, or complete metric reconstruction from every Ricci-carrier solution;
- complex-frequency poles, mode stability, quasinormal modes, or ringdown waveforms;
- a second-order physical-process first law or nonlinear horizon dynamics;
- asymptotically flat conformal-gravity scattering with a complete boundary phase space, metric/tetrad falloffs, and finite null-infinity flux;
- Hawking states, particles, a quantum BRST quotient, positivity, or unitarity.

In particular, the paper withdraws the broad phrase "no local causal truncation." Zero Cauchy data for the Ricci carrier is a local linear truncation. What is established is failure of the tested endpoint diagnostics to imply that truncation.

13 Certificate and reproduction ledger

The manuscript is an OUTLINE-ALLOWED/DRAFT-ALLOWED integration of committed certificates at repository baseline recorded in its generated claim map.

Claim	Authoritative certificate
Static spherical family	`black_hole_programme/certificates/BH0_STATIC_SPHERICAL_BACKGROUND.json`
Normalized generator and first law	`black_hole_programme/certificates/BH1A_NORMALIZED_GENERATOR.json`
Linear spherical dynamical extension	`black_hole_programme/certificates/BH1B_DYNAMICAL_EXTENSION.json`
Axial operator and Ricci–Bach composition	`black_hole_programme/certificates/BH2A_AXIAL_OPERATOR.json`
Extra horizon reach	`black_hole_programme/certificates/BH2A_HORIZON_REACH.json`
Action-derived axial current	`black_hole_programme/certificates/BH2A_FLUX_MATRIX.json`
Mixed and extra horizon flux fixtures	`black_hole_programme/certificates/BH2A_CROSS_FLUX.json`
Axial causal disposition	`black_hole_programme/certificates/BH2A_CAUSAL_DISPOSITION.json`
Polar trace-coupled split	`black_hole_programme/certificates/BH2B_POLAR_SPLIT.json`

Every certificate records a schema, exact or controlled-numerical payload, independent verifier, mutations or controls, command receipt, dependency tags, and missing claims. This manuscript must not advance to THEOREM-FROZEN until the static certificate's complementary-sheet wording is repaired, the flux fixtures acquire rigorous nonvanishing bounds, and the asymptotic Jordan/metric analysis closes. The polar scope remains a separate parity-completion gate.

14 Outlook

The next calculation is not a generic stability claim. The principal gate is the repeated-root asymptotic problem:

$$\text{first-order matrix and Jordan form} \longrightarrow \text{metric reconstruction} \longrightarrow \text{finite-flux boundary class.}$$

In parallel, one rigorous interval or symbolic horizon-pairing fixture should replace the present controlled numerical nonvanishing. The polar sequence then runs from the Zerilli control through horizon reach, Lee–Wald pairing, and the same outer audit. Only after those steps should complex-frequency structure and quasinormal boundary conditions be activated. A later asymptotic phase-space construction must decide which generator is physical, which flux reaches null infinity, and whether the additional curvature image survives a globally differentiable charge algebra.

The present result already fixes an important baseline:

> The axial non-Einstein curvature sector contains future-horizon-regular solutions and has nonzero controlled Lee–Wald pairing fixtures. Horizon analyticity and the presently tested leading asymptotic diagnostics do not select the Einstein kernel.

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